Chapter 11

OSCILLOSCOPE

Physical phenomena that occur in everyday life can be measured using measuring instruments. In today's increasingly advanced era, measuring instruments are more commonly found in the form of electrical measuring devices. These electrical instruments function by converting certain physical inputs into electrical quantities that can be read. One such electrical measuring instrument is the oscilloscope.

In general, an oscilloscope functions as an instrument for measuring voltage, frequency, and phase, and it is widely used in various fields. One such application is in the field of medicine, where the oscilloscope is used as a component of the electrocardiogram (ECG), an instrument for detecting a person's heartbeat.

11.1 Objectives

1. Students are able to understand and identify the functions and working principles of an oscilloscope.
2. Students are able to measure voltage, frequency, and phase difference from a voltage source.
3. Students are able to compare the frequencies of two AC voltages.

11.2 Tools and Materials

1. GOS-620FG Oscilloscope
2. Several connector cables
3. Series circuit of battery with one potentiometer

4. External function generator AG-202a (1 piece)

11.3 Theoretical Foundation

The oscilloscope is an instrument used to measure electric voltage, along with its frequency and phase, while simultaneously displaying the waveform of the measured voltage. The signal shape measured by the oscilloscope appears as a graph of voltage versus time, which essentially makes the oscilloscope function as a plotter. The signal forms displayed by the oscilloscope as a plotter can be compared either with respect to time (single trace) or with another signal (dual trace).
The oscilloscope has a main component called the Cathode Ray Tube (CRT), which serves as the place where the cathode rays are generated. The CRT consists of several parts: the electron gun, deflection plates, phosphorescent screen, and the glass enclosure. When in use, the cathode is heated, triggering the release of electrons, which are then accelerated toward the anode due to the voltage difference between the cathode and anode. The emitted electrons are projected from the electron gun as an electron beam toward the phosphorescent screen. This process is what gives the oscilloscope it is also referred to as the Cathode-Ray Oscilloscope (CRO).

During its path, the electron beam, before appearing on the fluorescent screen, will encounter two pairs of deflection plates: a pair of vertical deflection plates and a pair of horizontal deflection plates, made of metal and supplied with voltage. The voltage applied to the deflection plates generates an electric field that can deflect the direction of the electron beam.

The vertical deflection plates are connected to the vertical amplifier, which is linked to the signal input path, so the vertical deflection of the electron beam follows the waveform

of the input signal and can be adjusted using the VOLT/DIV button.
The horizontal deflection plates are connected to a horizontal amplifier, which is linked to the time base generator, also known as the sweep generator. The sweep generator produces a sawtooth-shaped signal, causing the voltage difference between the horizontal deflection plates to increase linearly, then drop to zero, and rise linearly again. This signal causes the electron beam to '*sweep*' across the screen from one end to the other, with the sweep rate adjustable using the TIME/DIV button.

11.4 Experimental Methods

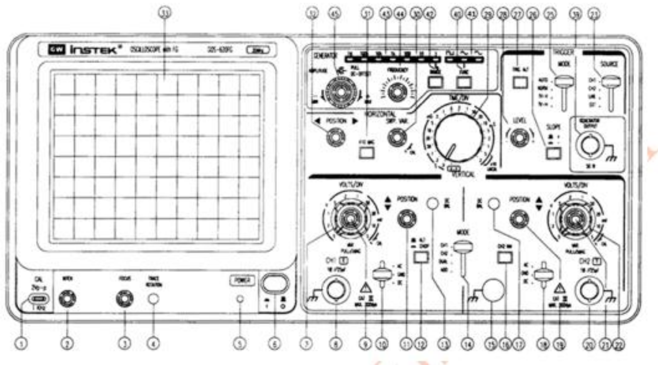


Figure 11.1: Front panel of the GW-INSTEK GOS-620FG oscilloscope

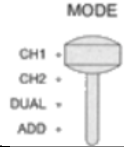
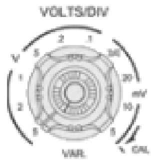
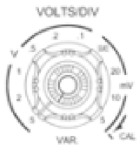
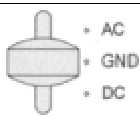
The Cathode Ray Oscilloscope (CRO or simply scope) is an electronic instrument widely used in various fields. One of the main advantages of the scope over a voltmeter lies in the fact that it does not contain any mechanical moving parts (such as needles, coils, damping systems, or springs); instead, all movements are carried out by a cathode ray (i.e., electrons), which practically have no mass. For example, events occurring within just a millionth of a second can be observed using a scope.

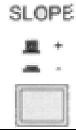
The oscilloscope is not only capable of measuring the magnitude of a voltage, but also of displaying the waveform and behavior of the voltage signal (the terms wave and vibration are often used interchangeably). In this experiment, you will learn how to operate the oscilloscope. For that purpose, you will need a basic understanding of how the oscilloscope works. Knowing its internal components will help you operate the control buttons on the front panel more effectively.

Setting Default

Before the device is powered on, the control knobs on the front panel should be set as follows (steps (a) through (r)).

Step	Button Function	No	Figure	Setting
(a)	Power button	6		OFF
(b)	Knop Inten	2		Middle
(c)	Knop FOCUS	3		Middle

(d)	Switch Mode Vert	14		CH1
(e)	Switch ALT CHOP	12		ALT
(f)	Switch CH2 INV	16		OFF
(g)	Knop vertical position CH1/CH2	11 and 19		Middle
(h)	Knop VOLTS/DIV CH1/CH2	7 and 22		50 mV/DIV
(i)	Knop VARIABLE CH1/CH2	9 and 21		Turn fully clockwise (CAL)
(j)	Switch coupling CH1/CH2	10 and 18		GND
(k)	Switch Trigger SOURCE CH1/CH2	23		CH1

(l)	Switch Trigger SLOPE	26		Positive
(m)	Switch TRIG ALT	27		OFF
(n)	Switch TRIG MODE	25		AUTO
(o)	Knop TIME/DIV	29		0.5 S/DIV
(p)	Knop SWP.VAR horizontal	30		CAL (far right)
(q)	Knop POSITION horizontal	32		Middle
(r)	Button 10x MAX	31		OFF

EXPERIMENT A:
OBTAINING A SHARP SPOT

CRO ACTIVATED

1. Make sure the CRO is set to its default settings (steps (a) to (r)). The POWER LED (number 5) will light up.

Wait approximately 10 seconds, and a spot will appear on the screen slowly moving to the right. If nothing appears, ask the assistant. Adjust the INTEN and FOCUS knobs to obtain a sharp spot. Set the INTEN knob moderately, not too bright.

2. Turn the INTEN knob (19) clockwise (CW) and counterclockwise (ACW). This adjusts one of the voltages directed to the electron gun, which determines the number of electrons fired toward the screen. If the knob is turned too far counterclockwise (ACW), all electrons will be blocked, and none will reach the screen. The spot on the screen will disappear (become invisible).
3. Turn the FOCUS adjustment knob (3) clockwise (CW) and counterclockwise (ACW), and observe the behavior of the spot. Finally, adjust the knob until the spot becomes as sharp as possible.

MAKING A STATIONARY SPOT

4. Turn the TIME/DIV knob (29) fully to the left, to the X-Y mode. You will see that the spot immediately becomes stationary (if the spot disappears, report to the assistant). If you have previously completed steps (g) and (q) regarding the POSITION properly, the spot will be approximately at the center of the screen (if not, leave it as is, as long as it does not disappear).
5. In the X-Y setting, the POSITION CH1 knob (number 11) no longer functions. The vertical position adjustment is done using the POSITION CH2 knob (number 19).
6. If you feel it is necessary, you may repeat the experiment with the FOCUS and INTEN knobs, while the spot is stationary. Now you must be careful, the

spot should not be too bright, just faintly visible, because it is stationary. The shape of the spot, in this stationary condition, may not be as good as it was when it was moving. This is caused by an imperfect ground connection. For the time being, do not worry if the shape of the spot is not perfect.

7. After you have observed all of this, turn the INTEN knob (number 2) counterclockwise until the spot just disappears. Do not turn off the oscilloscope.

WARNING!

If a sharp spot is continuously left in one fixed position (meaning the electron beam continuously strikes the same point) for a long enough period and with sufficient intensity, the fluorescent layer on the screen will burn, and at that point, the layer will be damaged and can no longer fluoresce. The oscilloscope tube will become defective or damaged.

- Therefore, when using the scope, do not turn the INTEN knob too high. Use only moderate brightness, just enough to display the image clearly.
- Turn the INTEN knob far to the left (ACW) if you are pausing from using the scope.
- Follow the instruction above: slowly turn the INTEN knob (2) to the left and observe the brightness of the spot until it disappears. Then, turn it back to the right until the spot just barely becomes visible again.

EXPERIMENT B:
MOVING THE SPOT ACROSS THE SCREEN

We have already managed to produce a sharp spot on the screen. But this is only the first step. Our ultimate goal is to obtain curves, graphs, or lines on the screen. To achieve this, the spot must be moved across the screen. How can this be done? (Remember, a luminous spot that moves fast enough gives the impression to the human eye of a continuous line).

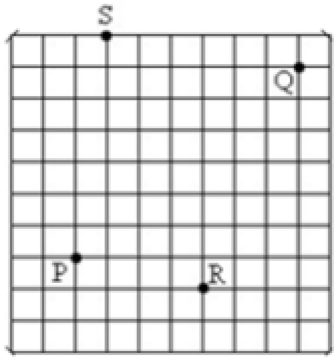


Figure 11.2: Place the spot at locations P, Q, R, and S

1. Your last action was to reduce the INTEN of the dot until it disappeared, and the position of the TIME/DIV (29) knob is on the far left, X-Y. Now make the dot appear again on the screen by turning the INTEN knob CW. Be careful not to make it too bright. Focus it as best as possible.
2. Turn (adjust) knob (19) and (32) so that the dot is in the center of the screen.
3. Place the dot at positions P, Q, R, S as in Figure 11.2.
4. Place the dot back in the center of the screen.
5. Reduce INTEN until the dot disappears exactly from the screen, do not turn off the scope.

SCOPE AS A GRAPH PLOTTER

If we make a graph that illustrates a physical event, then it is usually the case that the time quantity is placed on the horizontal axis or X-axis, with units such as years, seconds, microseconds, and so on. The faster the event occurs, the shorter the unit on the time axis. The scale on the time axis is linear, each cm on the graph paper represents the same time range, for example, 1 cm = 1 year or 1 $cm = 1$ µs.
Likewise on the scope screen, the horizontal axis is used as the time axis. Inside the scope box, there is what is called the time base circuit, which causes the dot to move from the left to the right across the screen at constant speed. When the dot reaches the right edge of the screen, it quickly jumps back to the left, then moves again from left to right in a regular manner. During the fly-back event, the brightness of the dot is turned off, so the path during the return is not visible on the screen.
If you observe the numbers, you will see that 0.5 S/DIV is the lowest setting and 0.2 µS/DIV is the highest setting

Note: ".5 S" means 0.5 seconds and ".2 µS" means 0.2 microseconds.

What does this mean? At the lowest setting, the dot spans (covers) one division in 0.5 seconds. The screen is 10 divisions wide, so at this setting, the dot takes 5 seconds to cover the entire width of the screen. In other words, each division on the horizontal time axis represents 0.5 seconds, thus 10 consecutive divisions represent 5 seconds. For the ".2 µS" setting, the dot moves through one division in 0.2 microseconds, so it spans the entire screen in 2 microseconds, two millionths of a second, an extremely short time span. We will verify this fact ourselves.

EXPERIMENT C:
USING THE TIME/DIV SWITCH

1. Adjust the INTEN knob until the dot appears on the screen.
2. Release the TIME/DIV knob from the X-Y setting, and set it to 0.5 S/DIV. A dot will appear on the screen moving to the right, from the left edge to the right edge. (If the disappearing dot does not reach the right edge before starting again from the left, it is POSSIBLE that you did not complete step 4 in the previous experiment, which is placing the dot in the center of the screen in the X-Y setting. If this happens, repeat step 4.) Next, adjust the moving dot so that it is centered vertically by adjusting (turning CW or ACW) the POSITION CH1 knob (number 11).
3. Turn the TIME/DIV selector switch, starting from the last position, which is 0.5 S/DIV, step by step to the right. Observe what happens on the screen, you will see that the dot moves faster and faster.
4. Measure the time it takes for the dot to move across 10 divisions (i.e., the full width of the screen = 10 DIV) from left to right for the following settings of the TIME/DIV selector switch (29).

Position TIME/DIVE (second)	Sweep time (second)
.5 s	……………………………
.2 s	……………………………
.1 s	……………………………
50 ms	1/…………………………
10 ms	1/…………………………
1 ms	1/…………………………
50 μs	1/…………………………
2 μs	1/…………………………
.2 μs	1/…………………………

5. Return the TIME/DIV selector switch (29) to ".5 S/DIV", and observe the dot moving slowly.
6. Reduce the intensity until the dot disappears; do not turn off the scope.

EXPERIMENT D: MEASURING DC VOLTAGE WITH A SCOPE

1. Increase the intensity just enough, not too bright. A dot moving slowly to the right will appear on the screen. Set the TIME/DIV switch to 1 mS/DIV. The dot will now appear as a line. At the 1 mS/DIV setting, the dot moves from left to right with a sweep time of 1/1000 second. This motion is too fast for the eye to follow, so it appears as a horizontal line. Use the POSITION CH1 knob (number 11) to place the line in the center of the vertical axis of the screen.
2. DC voltage will be measured using CH1. The DC voltage source used is a circuit of 4 batteries, each with a specification of 1.5V. If all the batteries are still new, the total voltage is approximately 6 volts.
3. Use two cables, red for + and black for –. Connect the red cable to the CH1 input (number 8) and to the positive terminal of the battery pack. Plug the black cable into the negative terminal of the battery and to the GND (number 15) of the oscilloscope.
4. Set the VOLTS/DIV CH1 switch (number 7) to the far left (5 V/DIV). The small gray rotary knob above the VOLTS/DIV switch MUST be all the way to the right (CAL). If the knob can still be turned, rotate it to the right (CW) until a "click" is heard. This CAL knob is used to calibrate the voltage reading. It does not need to be used if the voltage measured by the scope is already (still) accurate.

5. Set the CH1 COUPLING switch (number 10) to GND, this way, the direct voltage from the source is not yet connected to the scope.
6. Then set the same CH1 COUPLING switch to AC (up position).
7. You will notice that there is no change in the dot, because with the AC setting, ONLY ALTERNATING VOLTAGE IS PASSED THROUGH.
8. Now move the toggle switch to DC (down position). The line will deflect upwards. The amount of deflection depends on the DC voltage being measured and the VOLTS/DIV setting.
9. Complete the following section for a DC voltage source consisting of 4 batteries connected in series.

Setting VOLTS/DIV	DIV	Measured Voltage (volts)
5 VOLTS/DIV	....	....
2 VOLTS/DIV	....	....
1 VOLTS/DIV	....	....

DIV refers to the number of divisions, which can be read from the scale on the screen. The screen is divided into 10 large box sections (10 DIV), both vertically and horizontally. Each box is subdivided into 5 parts. So, the smallest screen scale is 0.2 DIV. If, for example, the line deflects upward by 1.2 DIV (from the vertical center axis), and the VOLTS/DIV setting is 5 V/DIV, then the measured DC voltage is 1.2 DIV multiplied by 5 V/DIV, which equals 6 V, this is the measured DC voltage.

10. You may observe that at the 1 V/DIV setting, the line disappears. Why? Because the screen cannot display a line that deflects by 6 DIV. From the vertical center axis (called the reference point), the maximum deflection is 5 DIV.
    How can the line appear on the screen if the voltage is 6 V and the setting is 1 V/DIV? The solution is: when COUPLING CH1 is set to GND, the line is adjusted not to the vertical center, but to the very bottom, for example, by adjusting VERTICAL POSITION CH1 (number 11). Perform this step.
    At the 1 V/DIV setting, how many DIV does the line deflect from the bottom line?
11. Also perform DC voltage measurements for 1 battery, 2 batteries, and 3 batteries using different VOLTS/DIV settings. Fill in the following table.

Number of batteries	Setting VOLTS/DIV	DIV	Measured voltage (volts)
1 battery	……….. ……….. ……….. ………..	……….. ……….. ……….. ………..	……….. ……….. ……….. ………..
2 batteries	……….. ……….. ……….. ………..	……….. ……….. ……….. ………..	……….. ……….. ……….. ………..
3 batteries	……….. ……….. ……….. ………..	……….. ……….. ……….. ………..	……….. ……….. ……….. ………..

12. Disconnect all cables. Reduce the INTEN until the line disappears. Do not turn off the scope.

EXPERIMENT E:
DISPLAYING AC VOLTAGE AND FREQUENCY

For this experiment, we will use the built-in FG (Function Generator) on the GOS-620FG oscilloscope we are currently using. The output from the FG is an AC voltage whose waveform can be selected, namely: square, triangular, and sinusoidal voltages. The control knobs of the FG are shown in Figure 11.3.

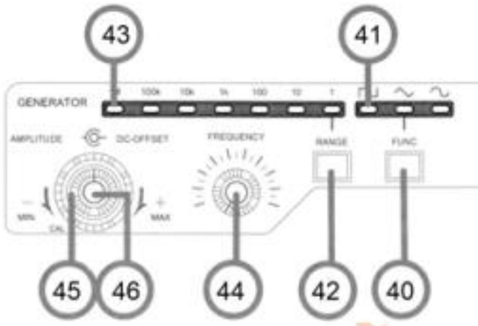
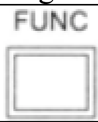
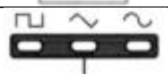
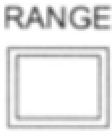
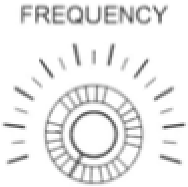
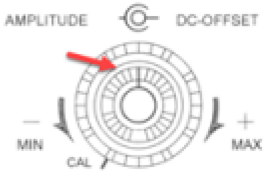
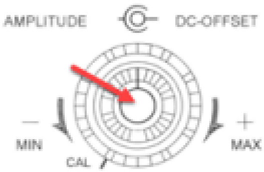


Figure 11.3: FG Control Knop Group

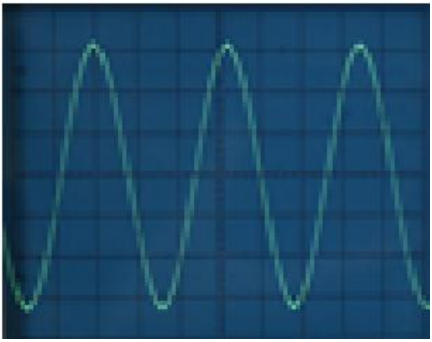
No	Figure	Function
40		Changing the waveform shape
41		Waveform shape selection indicator LED

42		Frequency RANGE selector switch, from 1 MHz, 100 kHz, 10 kHz, 1 kHz, 100 Hz, 10 Hz, and 1 Hz.
43		Frequency RANGE selection indicator LED
44		Fine frequency adjustment
45		Amplitude adjustment
46		DC-OFFSET

MEASURING PEAK-TO-PEAK VOLTAGE (Vpp)

1. Rotate the INTEN knob clockwise until a line appears on the screen.
2. Set the TIME/DIV selector switch to 0.2 ms/DIV.
3. Set the VOLTS/DIV CH1 selector switch to 1 V/DIV.

4. Press the FUNC button until the LED indicator for the sine wave is lit, meaning we are selecting the FG output in the form of a sine wave.
5. Press the RANGE button until the 10k LED is lit.
6. Connect the FG output to CH1 input.
7. Adjust the FG output wave amplitude so that the entire wave is visible on the screen, without the top or bottom being cut off. Remember that the VOLTS/DIV CH1 setting is at 1 V/DIV.
8. Adjust the FREQUENCY knob so that approximately 3 full waveforms are visible on the screen (it does not have to be exactly 3).
9. Draw the waveform visible on the oscilloscope screen (approximately like the image below).



10. (Fill in the blanks) Peak-to-peak deviation (in DIV) = ... DIV. VOLTS/DIV setting = ..., therefore the measured AC voltage = ... volts.

MEASURING THE FREQUENCY OF AC VOLTAGE

1. Turn the FREQUENCY knob to the far left.

2. Set the COUPLING CH1 switch to GND. Adjust the VERTICAL POSITION CH1 so that the line is at the vertical center. Then return the COUPLING to AC.
3. Adjust the amplitude knob so that the wave peaks (minimum and maximum) touch the grid lines on the DIV.
4. Measure the number of DIV horizontally for one wave. If necessary, use the HORIZONTAL POSITION knob to adjust the beginning of the wave to a readable DIV scale. Calculate the time for one wave, and calculate the frequency.

One wave (DIV)	Setting TIME/DIV	Period (second)	Frequency (Hz)
……	………..	…………..	…………..

5. Turn the FREQUENCY knob to the far right. Many waves appear on the screen.
6. Set the TIME/DIV switch so that one full wave appears on the screen.
7. Measure the number of DIV horizontally for one wave. If necessary, use the HORIZONTAL POSITION knob to adjust the beginning of the wave to a readable DIV scale. Calculate the time for one wave, and calculate the frequency.

One wave (DIV)	Setting TIME/DIV	Period (second)	Frequency (Hz)
……	…………	…………..	…………..

8. Turn the INTEN knob counterclockwise until the display on the screen disappears. Turn off the oscilloscope.
9. Tidy up the equipment used, disconnect the cables, and organize everything properly.